COMP448 Medical Image Analysis Homework 3 Report

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Part 1 - Network Construction

1. Optimizer Selection:

The Adam optimizer and SGD with momentum were compared by running each for 10 epochs. With SGD and momentum, the loss was decreased more slowly compared to Adam, resulting in a lower F1-score of around 0.40. F1-score of 0.60 was achieved with Adam in the first 10 epochs. Thus, Adam was selected as the optimizer.

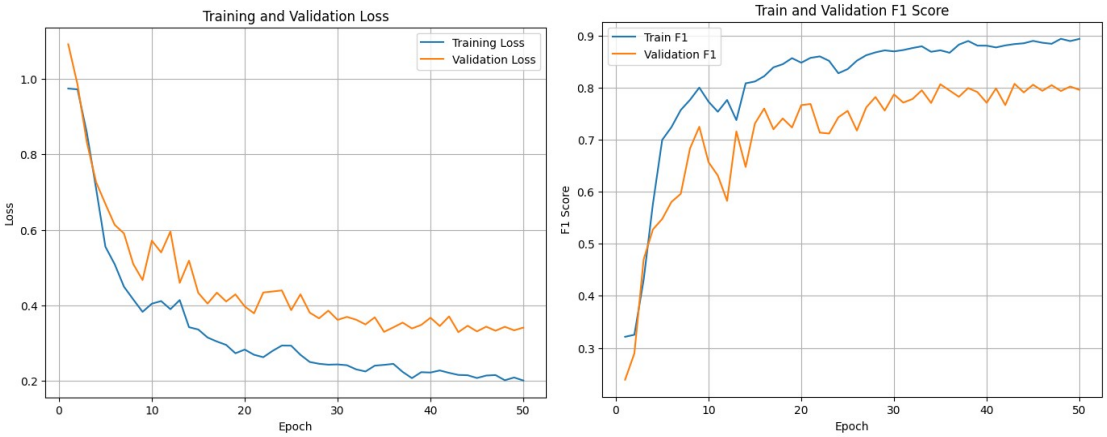
2. Batch Size:

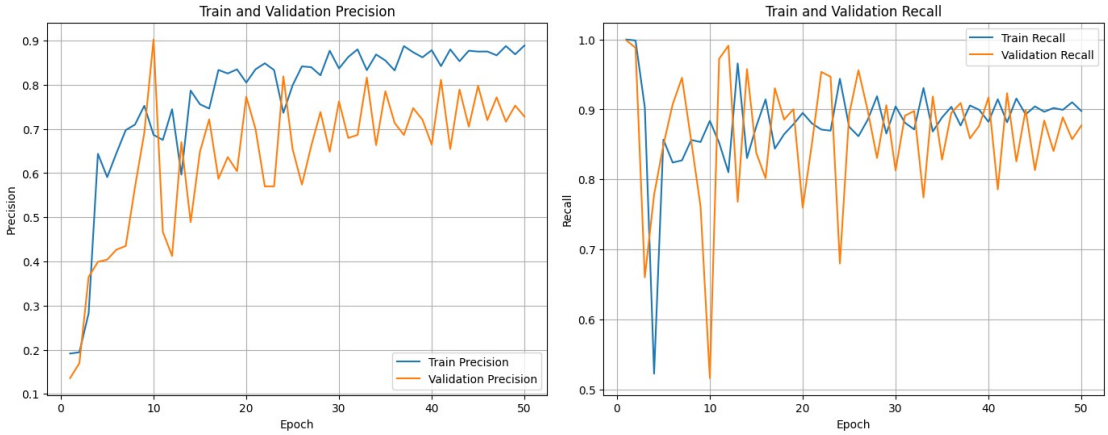
A batch size of 32 was used during training. Smaller batch sizes such as 8 were observed to increase gradient noise, and using 16 resulted in slower training compared to 32. Therefore, a batch size of 32 was chosen, as it resulted in more stable training and smoother loss curves, whereas batch sizes of 8 and 16 resulted in spikier loss graphs.

3. Early Stopping Condition:

Early stopping was based on validation F1-score to prevent overfitting and evaluate the performance with a comprehensive metric. After each validation epoch, the F1-score was computed and compared to the highest value until that point. So the F1-score is replaced every time there is a better F1-score, and the patience counter is set to 0. The model's weight with the best F1-score is saved every time, and the last saved model is used for testing. The early stopping condition is set to 20, so that training will stop if F1-score does not improve for 20 consecutive epochs. Additionally, A ReduceLROnPlateau scheduler with a patience of 5 epochs was used to avoid plateaus.

4. Training and Validation Performance Metrics





5. **Overall Performance**

F1 Score: 0.8373

Precision: 0.8066

Recall: 0.9023

Image Based Evaluation

<u>Good Segmentation Results</u>

Image 7

F1: 0.9659, Precision: 0.9531, Recall: 0.9790

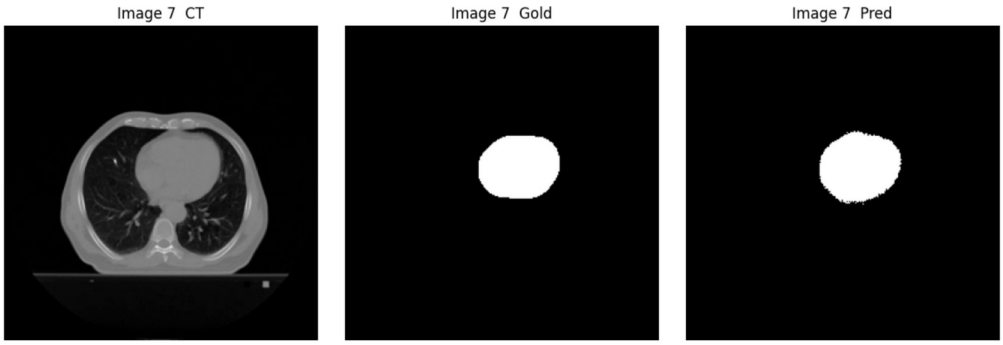
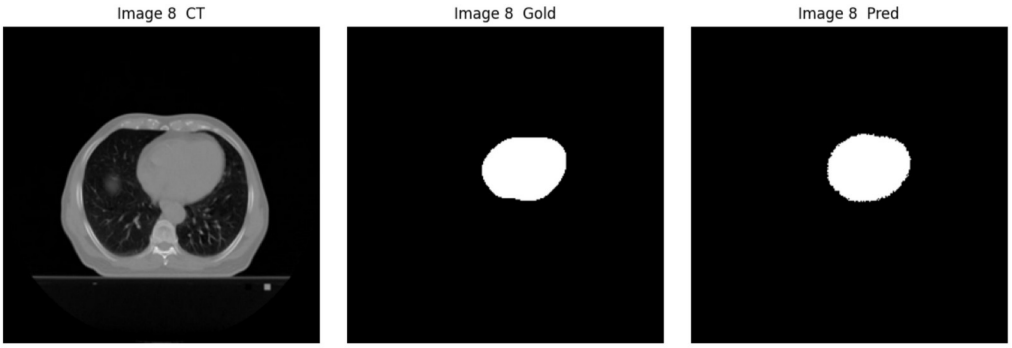


Image 8

F1: 0.9651, Precision: 0.9665, Recall: 0.9637



Acceptable Segmentation Results

Image 60

F1: 0.9146, Precision: 0.8877, Recall: 0.9432

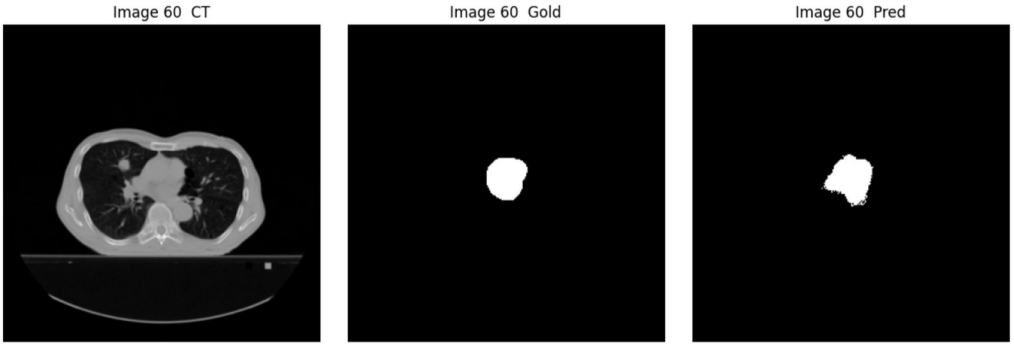
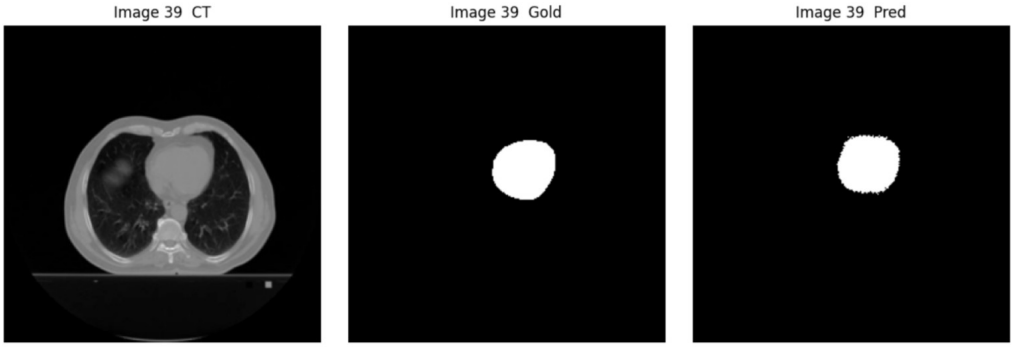


Image 39

F1: 0.9128, Precision: 0.9041, Recall: 0.9217



Bad Segmentation Results

Image 45

F1: 0.1788, Precision: 0.1041, Recall: 0.6347

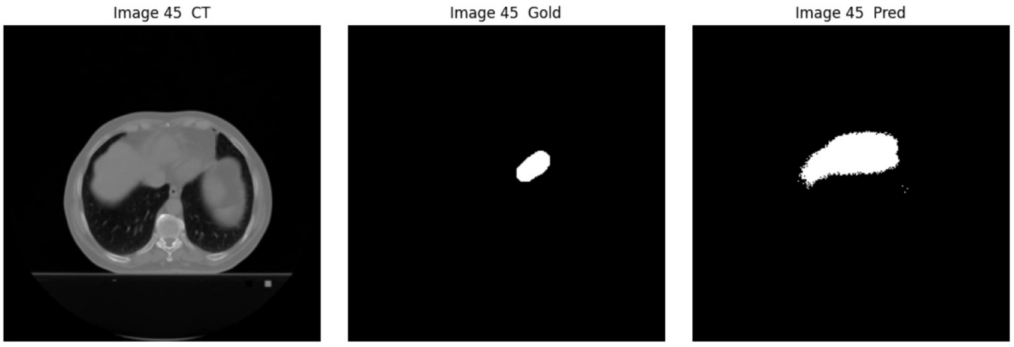
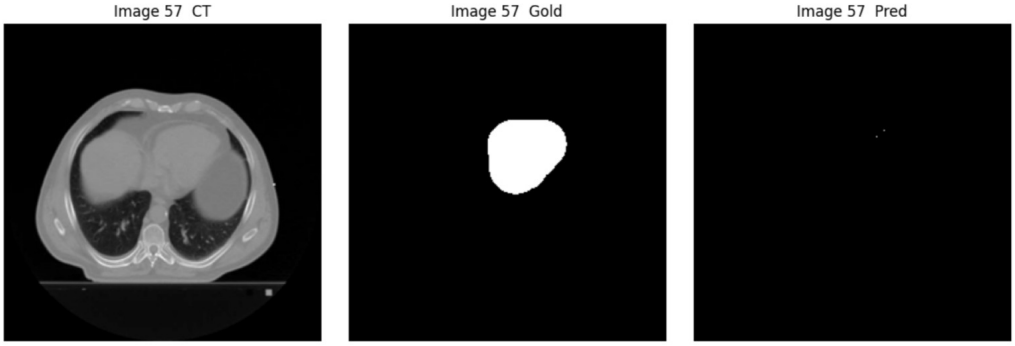


Image 57
F1: 0.0000, Precision: 0.0000, Recall: 0.0000



Based on these segmentation results, it can be seen that for good segmentations, the model's prediction matches the gold standard, indicating the success in learning spatial boundaries and shapes. In acceptable segmentations, the core structure is segmented, but there are problems in boundary refinement. In problematic segmentations, the model either oversegments or completely fails. It can be concluded that the model performs best in segmenting the edges and boundaries of round or well-defined shapes.

Part 2 - Network Architecture Modification

	Training Set			Validation Set			Test Set		
	Precison	Recall	F-score	Precison	Recall	F-score	Precison	Recall	F-score
Default network design	0.8026	0.8908	0.8542	0.7516	0.8649	0.8043	0.7973	0.9217	0.7973
Modification in the number of down and up sampling steps	0.6625	0.9180	0.7696	0.4805	0.9142	0.6299	0.8457	0.8922	0.8478
Modification in the number of feature channels	0.7430 (8) 0.8653 (32)	0.8961 (8) 0.8745 (32)	0.8124 (8) 0.8699 (32)	0.6410 (8) 0.7571 (32)	0.8570 (8) 0.8256 (32)	0.7334 (8) 0.7899 (32)	0.7845 (8) 0.8227 (32)	0.8668 (8) 0.8930 (32)	0.8029 (8) 0.8271 (32)

1. **Which method performs the best segmentation performance over other methods? What are the possible reasons for this difference?**
   The modification in the number of down and up sampling steps performs the best in terms of test F-score (0.8478). Both modifications outperformed the default network design. This performance difference can be because of the better generalization when network depth is reduced. Additionally, fewer downsampling steps may have preserved more spatial details.

2. **What are the observed differences over the default network design in terms of segmentation performance, training time, and the gap between training and test set results of methods?**

   Segmentation performance: Both architectural modifications resulted in higher test scores compared to the default model. The model with reduced downsampling achieved the highest F1-score of 0.8478, followed by the model with increased feature channels at 0.8271, and the model with reduced feature channels at 0.8029. However, the model with reduced downsampling showed poor validation performance, suggesting overfitting.

   Training Time: In terms of training time, the 8-channel model was the fastest to train, considering its lightweight architecture and reduced complexity. The reduced downsampling model also trained slightly faster than the default model. The 32-channel model was the slowest among all models, as the increased number of channels resulted in longer epoch durations.

   Gap between Training and Test Results: The gap between training and test results was most consistent in the model using 8 feature channels, indicating better generalization and less overfitting. The default network design performed well across all sets. In contrast, the model with reduced downsampling/upsampling had a notable gap between training and test result. It performed well during training but the performance dropped on test metrics, indicating that it struggled to generalize on unseen data.

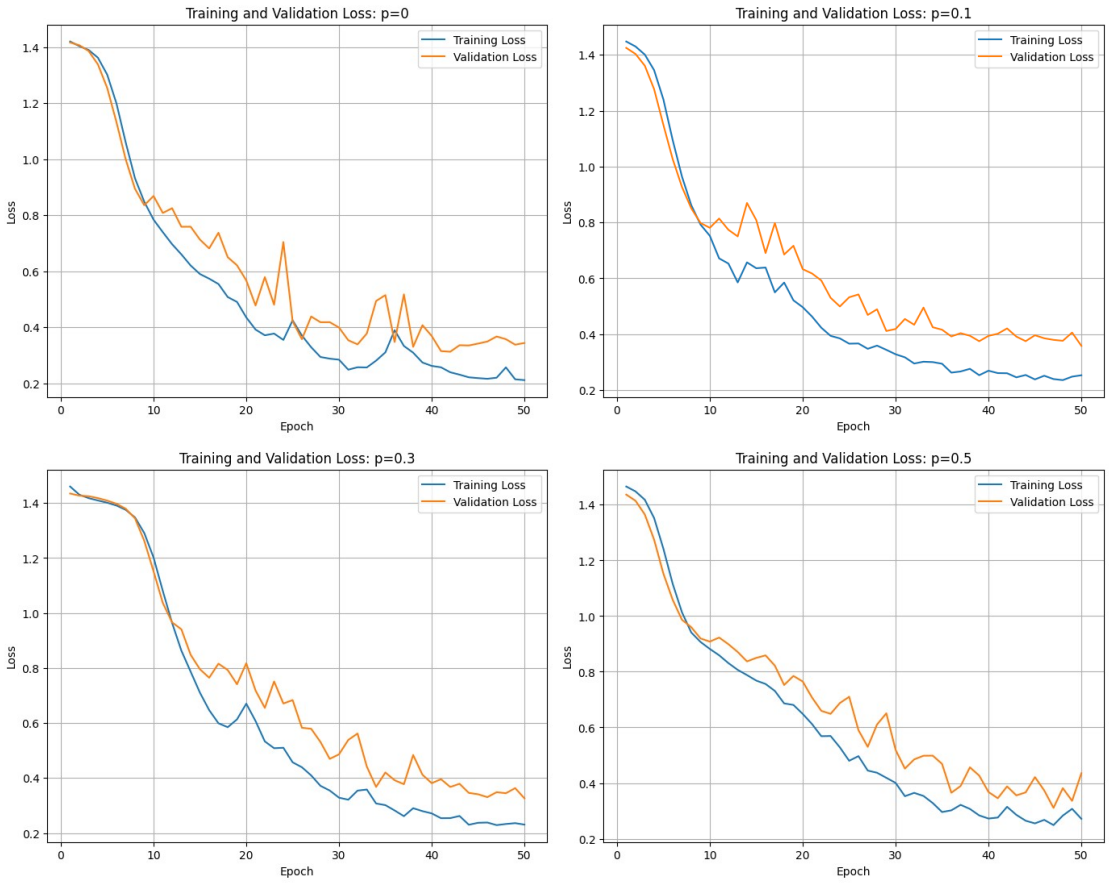
3. **Which network architecture do you prefer among these three methods and why?**
   Considering the performances on training, validation, and tests, I would prefer the model with 32 feature channels since it is the most balanced and reliable model. While the model with modified down and up sampling steps has the highest test F1-score, it performed poorly on the validation set, suggesting instability on certain data. In contrast, the model with 32 feature channels performed consistently across all sets with also high precision and recall scores. This indicates better generalization and lower risk of overfitting.

Part 3 - Dropout in Network Architecture

The dropout layer is added to the bottleneck, after the deepest encoder block and before the first decoder block. We decided to do that since the bottleneck has the highest number of feature channels, representing high-level information. Adding dropout to the bottleneck would allow the network to generalize better.

	Training Set			Validation Set			Test Set		
	Precision	Recall	F-score	Precision	Recall	F-score	Precision	Recall	F-score
Default network design	0.8504	0.9188	0.8833	0.6943	0.9149	0.7895	0.8124	0.9202	0.8387
Network w/dropout (p=0.1)	0.8235	0.9058	0.8627	0.7108	0.8626	0.7794	0.7892	0.8935	0.8235
Network w/dropout (p=0.3)	0.8489	0.8963	0.8719	0.7711	0.8551	0.8109	0.7853	0.9288	0.8347
Network w/dropout (p=0.5)	0.8424	0.8665	0.8543	0.5695	0.9584	0.7144	0.8046	0.9261	0.8326

1. **Do you observe the regularization effect in your results after integrating the dropout layers? What performance results demonstrate the regularization effect of dropout layers?**
   Based on the results, it can be seen that the train-test F1-score gap decreased after introducing dropout layers. This suggests that the model generalizes better after applying dropout. Even though the training performance is lower, the validation curves are smoother and validation-loss curves track training-loss more closely in the networks with dropout. It can be said that reduced train-validation F1 gap, smoother validation-loss curves, and the preserved test-set accuracy demonstrate the regularization effect of dropout.



2. **Which p-value performs the best segmentation performance over the other two p-values? Comment on the possible reasons for this situation.**

   When three dropout rates (0.1, 0.3, and 0.5) are compared, the best performing dropout rate is 0.3 with the highest validation (0.8109) and test scores (0.8347). This suggests that moderate level of dropout (between 0.1 and 0.5) provides enough regularization to prevent overfitting while still allowing the network key features. It is possible that p=0.1 is too weak to prevent overfitting, and p=0.5 causes underfitting.

3. **Could we say that by increasing the p-value in the dropout layers, the difference between training and test set performance results decreases depending on your results?**
   Train test gaps are 0.0446, 0.0392, 0.0372, and 0.0217 in p=0, p=0.1, p=0.3, and p=0.5 respectively. So it can be said that increasing the p-value decreases the train-test difference based on these results.